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Producing calculable worlds: education at a glance

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The OECD's international education indicators have become very influential in contemporary education policies. Although these indicators are now routinely, annually published in the form of *Education at a Glance*, the calculability upon which the indicators depend was an achievement that involved the mobilisation of a huge machinery of expertise, trust, pragmatism and other resources. This paper traces the ways in which varied constraints were addressed, interests translated, categories defined, classifications negotiated, frameworks agreed upon, choices made, methodologies established and protocols developed, as the indicators exercise moved from being nearly impossible to becoming routinely produced. Using resources from Science and Technology Studies (STS), it demonstrates that the work of making such assemblages is both instrumentalist and performative, and argues for an undertaking of critique as a moral enterprise.

Keywords: sociology of measurement; OECD; large-scale comparisons; education policy

In recent years, the Organisation for Economic Co-operation and Development (OECD) has become a key producer of education indicators that participate in policy enactments in many countries. These indicators seek to describe education systems not only precisely but also *comparatively* and *predictively*, with details about a host of parameters, such as the proportion of students expected to complete tertiary education; the differences in the career aspirations of boys and girls; spending on education; labour market outcomes; rates of transition from school to work; teaching time; and class size. The triennial Programme for International Student Assessment (PISA) surveys and their international rankings of student performance generates greater media attention; however, another publication, the now annual *Education at a Glance* (EAG), is a far more comprehensive document, and is deeply influential in policy circles:

The use of [EAG] indicators for comparisons and strategic mobilization is now a regular part of educational politics. For example, after comparing unfavorably among the 1996 EAG indicators of teacher salaries with their Belgian and German neighbors, Dutch teachers used these indicators to lobby for increases in their salaries. Similarly, in the United States, comparisons of nations across a statistical indicator of dropouts were used to highlight comparatively low high school completion rates in 2000. In an extreme, but illustrative, case, one nation's incumbent political party requested that the publication of EAG be delayed until after parliamentary elections because of the potentially damaging news about how its education system compared to other OECD nations. (Smith & Baker, 2001, p. 142)

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In part, EAG's influence derives from the vast amount of information it carries. Despite its name, EAG is anything but 'at a glance' – it is a long and serious gaze that 'examines the quality of learning outcomes, the policy levers and contextual factors that shape these outcomes, and the broader private and social returns that accrue to investments in education' (OECD, 2011, p. 3). As one OECD official involved in the publication of EAG at the OECD explained:

It's 30 indicators, it's 230 tables, 140 charts and more than 100,000 figures. ... It's 'at a glance' but we have 500 pages, 550 in English. ... And this is one of the flagships of the OECD. PISA is more famous because it concerns more countries and also all students pass the same exam, but in EAG you have very useful indicators on how the educational system all over the world works, and ... we have the outcome, we have the finance, we have the input as well as the output. (OECD official, interview transcript)

EAG is a remarkable document indeed, with comparable data for the 34 OECD member nations as well as several non-member nations. The OECD presents EAG as 'the authoritative source for accurate and relevant information on the state of education around the world' (OECD, 2012, back cover), and, further, that EAG's indicators address:

the needs of a range of users, from governments seeking to learn policy lessons to academics requiring data for further analysis to the general public wanting to monitor how its country's schools are progressing in producing world-class students. The publication examines the quality of learning outcomes, the policy levers and contextual factors that shape these outcomes, and the broader private and social returns that accrue to investments in education. (OECD, 2012, p. 3)

The insouciance and confidence with which the OECD routinely, annually, produces such a range of indicators to inform and influence governments, academics and the general public belies the decades of hesitation, experimentation, negotiation and effort that went into the development of these international education indicators. As Norberto Bottani, who headed the OECD's international education indicators project for several years from its inception, reflected:

When in 1988 the Organization for Economic Co-operation and Development (OECD) launched a project aiming to produce a set of international education indicators, only a minority of experts, statisticians, decision-makers, scholars and, last but not least, OECD colleagues believed that it would be possible to produce a reliable, robust and policy-relevant set of international education indicators. Six years later the OECD education indicators are internationally recognised as essential statistics, frequently quoted by the international press, widely used in debates on education policy in national parliaments and carefully scrutinised by civil servants within ministries. (Bottani, 1994, p. 333)

It is this transition from being a nearly impossible project to becoming 'essential statistics ... widely used in debates on education policy in national parliaments' that is the focus of this paper. Tracing how these indicators come to cohere and gain traction, I describe how varied constraints were addressed, interests translated, categories defined, classifications negotiated, frameworks agreed upon, choices made, methodologies established and protocols developed. It is a story of policy assemblage at a transnational, transdisciplinary scale, involving their increasing acceptance as neutral and scientific

representations; as artefacts of method, from which traces of the decisions and actions of the many actors involved have gradually been erased (Derksen, 2000).

This story is itself an assemblage, pieced together mainly from publications by Norberto Bottani, who was associated with the OECD from 1976 onwards in various capacities. With a Master's degree in Education from the Faculty of Philosophy, University of Fribourg, Bottani's initial work at the OECD was in early childhood education. From 1983 to 1987, he headed the Education, Cultural and Linguistic Pluralism project at the OECD. He was Head of the OECD Unit for Education Statistics and Indicators between 1994 and 1987, and Head of the Project for International Indicators of Education Systems (INES), the unit that was responsible for developing the international education indicators system which this paper is about, from its inception in 1988 through to 1997, which was just around the time that PISA was being conceptualised and developed. Throughout this time, Bottani published prolifically in OECD publications. Publications between 1992 and 1998, with titles such as *The OECD Education Indicators: Purposes, Limits and Production Processes* and (with OECD colleague Walberg) *Introduction: What are the International Education Indicators For?*, were optimistic about the advantages of having such indicators, but also careful to highlight their limitations. The general thrust was that, in the hands of those who understood their limitations, international indicators could provide valuable information. In 2008, by which time Bottani was no longer at the OECD and after EAG had become well established and routinely used, Bottani published *The oil level, the engine and the car; the stakes involved in assessing the quality of education on the basis of indicators* as one of the framing background papers for a conference on international comparisons in France in November 2008. That paper reflected his disillusionment with the indicators, and even more so with how they are being misunderstood and misused.

Other important sources of information have been the publications of Claude Sauvageot, who was the Vice-Chair of the INES Advisory Group between 2008 and 2011. He was the French representative to OECD's Centre for Research and Innovation in Education (CERI – INES grew out of CERI) from 2004 to 2011, and a member on the Standing Group for Indicators and Benchmarks of the European Commission. Two key publications by Sauvageot have informed this paper: *Indicators for Educational Planning: A Practical Guide*, published in 1977 by UNESCO, and a 2008 paper on the International Standard Classification of Education (ISCED) published in the special edition of *Education & Formations* to provide background to a conference on international indicators. The *Guide* was a result of his work in indicator development in a number of African nations, and provides advice to governments and indicator developers on what makes a good indicator and how to develop indicators. Other firsthand accounts that informed this paper were those of John Smyth, who was associated with UNESCO from 1972–2000, and of Stephen Heyneman who was with the World Bank from 1977 to 1998. Both Smyth and Heyneman wrote about the International Classification of Education (ISCED) which underpin the OECD international education indicators.

The paper is set out as follows: a brief introduction to Science and Technology Studies (STS) and 'sociology of measurement', which explains the theoretical approach used in this study is followed by an explanation of indicators and how they are different to statistics. After that, I tell the story of the OECD indicator development project. The section that follows analyses the changing roles of scientists – statisticians, education and

assessment experts, social scientists and psychometricians – as the project progressed. Further discussion follows about how ‘calculable worlds’ are produced. The paper concludes with an argument for the moral imperative of recognising and critiquing the performativity of contemporary policy enactments.

STS and ‘sociology of measurement’

The methodological and analytical approach in this paper is informed by STS (sometimes known as ‘Science Studies’ or ‘Science, Technology and Society’). From its origins in the early 1960s, STS has drawn together historians and philosophers of science, sociologists, anthropologists, economists and scholars in political and legal sciences. Seeing science not only as a human endeavour, but also as a social system, STS scholars reject dichotomies such as nature and science, and society and culture. The central concern of STS is to understand how knowledge is produced, mobilised and validated, how it circulates and how it comes to be challenged. This provides a very useful way to study governance modes such as ‘evidence-based’ policy making, since science and the state are seen as co-produced (Jasanoff, 2004). STS provides the tools by which we might:

explore how knowledge-making is incorporated into practices of state-making, or of governance more broadly, and, in reverse, how practices of governance influence the making and use of knowledge. States, we may say, are made of knowledge, just as knowledge is constituted by states. ... Knowledge, in particular, is seen as crystallising in certain ontological states – organizational, material, embodied – that become objects of study in their own right. (Jasanoff, 2004, p. 3)

The study of the interplay between the state and statistics has been the focus of some STS scholars, who have elaborated the how statistics became an invaluable and necessary means of governing large and complex provinces. Tracing the history of statistics, Desrosières (1998) explored ‘the politics of large numbers’, and described how the practices of scientists and the meanings attached to sciences change over time, and how these changes influenced, and were influenced by, the needs of state-making. Porter (1995) described how it is that numbers have come to be trusted and how ‘objectivity’ has come to be attached to numbers. Scott (1998) described how order and measurement make governance possible, and how the very act of ordering and measuring the world also changes the world. STS thus offers useful concepts with which to study the imbroglio of knowledge-making and state-making that has made the world of education calculable and comparable.

Latour’s work on tracing the practices of ‘science in action’ (Latour, 1987, 1999; Latour & Woolgar, 1979) guides my methodological approach in this study; I regard the development of international education indicators as *science in the making*, much as Latour approached the studies of scientists in laboratories. My focus is on the practices by which OECD’s international education indicators, once thought to be nearly impossible to produce, cohered as a viable assemblage and are now routinely and annually produced and widely used.

Woolgar’s ‘sociology of measurement technologies’ (Derksen, 2000; Gorur, 2014; Woolgar, 1991b) has been particularly useful to this study of the practices of *measurement*. Regarding measurement as a ‘technology’, Woolgar draws attention to the ‘instrumentalism’ of measurement practices (the ways in which things are ‘made to work’ in a pragmatic sense) and the use of instruments (in this case, surveys,

methodologies, calculations and models). Another crucial aspect of such a sociology is understanding measurements as not merely *descriptive*, but *productive* (Knorr Cetina, 1999) or *performative* (Law, 2009a; Pickering, 1995) i.e., they are ‘world making’ processes. As soon as the measurement exercise begins, it acts upon the world, changing priorities and influencing behaviours, policies and practices. In telling this abbreviated history of the OECD indicators, I focus on their *instrumentalism* as well as their *performativity*.

Indicators – more than mere ‘statistics’

Quantification and statistical mapping have been features in education since the 1850s, when education came to be recognised as a field of statistical inquiry. For many decades, the data collected included the numbers of schools, students enrolled and teachers employed. Sauvageot (1997) suggested that such statistical information was not very useful, because data were often not accurate, let alone timely or consistent, and there were too many details. Crucially, they did not include any *analysis*. What a busy policy maker needed, Sauvageot argued, were *indicators*:

A decision maker needs more analytical and relevant information, which is presented in a user-friendly way. The development of an *indicator system*, which is updated regularly, is essential if information is to be used properly. (Sauvageot, 1997, p. 3, my emphasis)

Indicators are different to mere statistics in significant ways. They are *selected statistics* specifically *designed to inform policy makers* about the state of the education system (Blank, 1993). They anticipate policy questions and provide information for policy decisions. They often combine data from multiple sources, including data specifically collected within certain regulated, purpose-built frameworks to facilitate comparison:

Uniquely different from the usual policy-related statistical analysis, statistical indicators are derived measures, often combining multiple data sources and several ‘statistics’ that are uniformly developed across nations, are repeated regularly over time, and have come to be accepted as summarizing the condition of an underlying complex process. (Smith & Baker, 2001, p. 141)

Indicators are thus *purposefully processed* according to assumptions about how education systems function, using mathematical models that inscribe the assumed relationships. They are not self-explanatory; rather, their meanings are connected with the situations within which the measures are constructed (Jaeger, 1978). They can be constructed to map and represent ‘the combined effect of several variables, including exogenous factors’ (Mainguet & Baye, 2006, p. 154). Indicators thus wrap up complex relations into a simple shorthand (Cullather, 2007).

The function of indicators is not to identify or assess problems, but to alert policy makers to them, and to help identify and measure the seriousness of the problem, rather like a warning signal on a dashboard. As Sauvageot (1997, p. 17) explained, ‘When the light goes on, a specialist has to find out why, and then find the solution to the problem’. Good indicators, according to Sauvageot, are constructed around the most important policy phenomena and policy objectives of an education system, and should describe education systems simply and precisely; provide a general overview; provide points of

comparison and other analysis, including trends over time; and report on diversities or disparities.

The making of the OECD education indicators

Established as an organisation focused on economics, there was initially no explicit mention of education as an area of concern for the OECD. However, an awareness of the relationship between social and economic factors, and the idea that education could contribute to economic growth, led to its involvement in education in the 1960s. In 1968, it set up the Centre for Educational Research and Innovation (CERI). In the 1970s and 1980s, CERI research focused on early childhood (Bottani headed the Early Childhood Project) and the transition of young people into the life of work. In the 1990s, amid growing discourses and practices of globalisation, CERI began to actively facilitate an exchange of views and information on education between countries.

The OECD's early efforts to develop international education indicators in the 1970s were daunted by the challenges of comparability and the lack of data, but interest in setting up international indicators emerged again strongly in the 1980s. In 1988, the International System of Education Indicators (INES) grew out of CERI to explore more vigorously the feasibility of developing a set of rigorous international education indicators for its member nations. Norberto Bottani, who headed the INES project from its inception for nearly a decade, reports that the purpose of the project was to 'gather, process, improve, and report statistics and indicators on education and related fields' (Bottani, 1996, p. 279). The project came to be seen as urgent following the sense of crisis in education that ensued after the publication of the influential *A Nation at Risk* (National Commission on Excellence in Education, 1983) in the USA. The type and quality of data available at the time were considered inadequate for comparing systems and improving performance (Bottani, 1996).

As member nations' desire for robust international indicators gathered force, a conference was organised by the US Department of Education and the OECD Secretariat in 1987 in Washington, DC, attended by representatives from 22 member nations, along with invited experts and observers from OECD nations (Bottani & Walberg, 1992). Two important points emerged from this conference: international benchmarks and comparisons had become 'an integral component of the improvement of the quality of education'; and there was a broad understanding that schooling needed to be improved, and data were needed to do that (Bottani & Walberg, 1992, p. 7).

Another conference soon followed at Poitiers, where 'the need for better and more comprehensive information about the outcomes of education' (Bottani & Walberg, 1992, p. 8) was emphasised. This meant going beyond the types of data (enrolment, number of teachers etc.) that were required for 'management' in an era of massive expansion of education provision, and towards a focus on data on the *quality* and the *outcomes* of education.

The main challenges facing INES were the lack of good statistical data, and the weak degree of comparability of the data that were available (Bottani, 1994, p. 334). In addition to these technical issues, INES had to grapple with a complex array of ambitions, priorities, fears and pressures:

[T]here were many sources of pressure aiming to stifle or modify the project right from the start. In the coalition of opponents, we find the following: education statisticians with

scientific claims condemning the indicators' imprecision and the biased view of education they convey; representatives of teacher associations with political arguments contesting the existence of the education quality crisis, particularly in the public sector; a wide range of teachers or specialists in the education sciences engaged in pedagogical innovations and progressive educational movements, invoking ethical arguments condemning the governments' intent to introduce new forms of education standards based on new public administration techniques, public sector governance and administration performance management. (Bottani, 2008, p. 16)

Early on, one of the big debates was which indicators to include, and how many indicators should be published. A wide range of indicators would address issues important to a number of countries. However, having too many indicators would make them difficult to navigate and more likely to be ignored. Not all countries produced the data that were needed for important indicators, and would need to invest funds to collect them for the purpose of developing new indicators. All this required negotiation between the users and the providers of data.

The recognition that the project could never be perfect was key to its feasibility. It was also clear that this enormous and challenging task could only be accomplished with a great deal of cooperation and goodwill. Several voluntary networks sprang up to assume different tasks in what has been described as 'one of the largest educational research efforts ever carried out':

By the completion of the exploratory and development phases of the indicators in 1991, several hundred policy makers, senior government staff and leading scholars had been involved. Measured by such involvement, the scope of the variables and the hundreds of millions of educational staff and students counted and surveyed, the OECD indicators project is undoubtedly one of the largest educational research efforts ever carried out (Walberg & Zhang, 1998, pp. 55–56)

As a result of this massive effort, a report was produced in 1991, and was discussed in a meeting in Lugano between the producers of the indicators and policy makers who would use the indicators. This meeting generated a great deal of debate, particularly around the inclusion of student attainment data. Given its uneven quality, statisticians were unwilling to include attainment data, but given their assumed importance, policy makers were keen to include them. Another challenge was deciding how many and which indicators to include and which ones to abandon.

The following year, the report was modified and published as the first EAG 1992, with some student attainment indicators included. EAG 1993 and 1995 followed. From then on, EAG has been published annually, and over time has grown to include a number of new indicators, as well as reporting on more education systems. No longer 'at a glance', the report produces indicators under following headings (OECD, 2012):

- A. The output of educational institutions and the impact of learning
- B. Financial and human resources invested in education
- C. Access to education, participation and progression
- D. The learning environment and the organisation of schools

The move from the initial tentativeness to the confident annual publication of EAG is a story of 'science in the making', involving many processes of 'ordering' (Berg & Timmermans, 2000; Bowker & Star, 2000; Law, 1994) to standardise a diverse world. In

the next section, I elaborate a small part of the fascinating story of how the OECD made the world calculable. The focus is on the changing fates of science and scientists – the measurement experts, the statisticians and the assessment experts – in the move from the early attempts at developing indicators in the 1970s, to the current routine, annual publication of EAG.

Science serving policy: sacrificing precision and purity

The theoretical and scientific challenges of the task of developing international indicators were well known to the OECD experts long before INES was set-up. The OECD had first begun developing education indicators in the 1970s. In 1973, it published *A System of Education Indicators to Guide Public Policy Decisions*, which attempted to measure the influence of education on social well-being (Bottani, 2008). With this very first attempt, the experts on the working group came up against a significant conceptual conundrum. In determining the facts that affected education and social well-being, ‘internal factors’ (relating to individuals) and ‘external factors’ (relating to macro or societal issues) were inextricably linked together and could not be meaningfully extricated for analysis. The dual role of these factors as both inputs and outputs confounded measurement – were they ‘parameters’ or ‘variables’? The response from the working group to these conundrums was to abandon the search for a workable theoretical model, and instead build the indicators around the notion of *usability*:

The group of experts responsible for drawing up this first OECD report on education indicators concluded by deciding to adopt a methodological approach which was unusual in scientific circles of the time, abandoning as it did the idea of developing a theoretical model of the education system. (Bottani, 2008, p. 14)

Several rationalisations were used to justify the adoption of this ‘unusual’ methodological approach such as constraints of time and ‘great political pressure on the group of experts to deliver an *operational* set of indicators’:

The group realised that it could not initiate lengthy preliminary discussions on a universal model of education system because an agreement would probably never be reached and so it opted for an empirical approach based on the study of national education-policy objectives. This led to developing a set of 46 indicators but these were never calculated, given that *the project’s fundamental scientific ambition was inconsistent with the available means and the interests of the government authorities which would have had to fund the initiative*. And so the social science specialists were unable to see their project through to its conclusion. It was a failure. (Bottani, 2008, p. 14, my emphasis)

The publication of *A Nation at Risk* in 1983 generated a panic about the quality of education systems, prompting renewed efforts to generate international indicators.

One of the challenges to developing a sound conceptual framework to underpin the choice of indicators is that a large number of factors influence educational outcomes (Bottani, 1998). Moreover, their impact depends on the presence of a host of other factors. Because it is not possible to measure all the factors that affect student attainment, it would be reasonable to include those factors which are considered relevant by most nations. But agreement on relevant factors was not easy to obtain. Therefore, the working

groups had to choose which indicators would be most useful, and '[t]hese choices are neither simple nor neutral' (Bottani, 1998, p. 62).

The OECD adopted a very simple input/output conceptual framework with a view to producing broad information about education systems which, whilst not suitable for detailed analysis of relationships between resource investment and results, would nevertheless serve to 'raise certain relevant questions at the macro-economic level' (Bottani, 1998, p. 63).

So from the beginning, INES regarded this project as one where science was in the service of policy. Rather than fuss about precision and purity of methodology, a pragmatic approach was taken, using the rationale that imperfect data were better than no data at all. Indeed, an insistence on 'statistical purity' was seen as obstructionist:

The lack of truly comparative data has been used as a block to the development of indicators and to question their validity and reliability. ... In developing a set of education indicators, it is ... *important to refrain from applying an overly rigorous test for the validity of the indicators*, and rejecting provisional measures or organising schemes because they do not account for all of the complexities of education and society. *The point is that indicators are not primarily designed for scientific research*. They can be useful in a more limited and practical way, even if they do not account fully for these complexities. (Bottani, 1994, p. 335, my emphasis)

Thus distinctions were made between a 'scientific research' enterprise on the one hand, and a 'practical' policy enterprise on another. At this point, not taking a 'research-oriented' approach was seen as sufficient, since the project was not intended to be used to make causal inferences or any detailed evaluations:

In order to produce policy-relevant variables that can be clearly and succinctly summarized, presented and considered, it was not necessary to adopt a complex and research-oriented model ... *The indicators were deliberately not organized according to a 'model'. A model would imply a sophisticated, causal connection among indicators; it is not reasonable to assume that the variables represented by these indicators function in these ways*. From the very beginning of the INES project there was consensus about the fact that the organization and the selection of the data was not compatible with a causal model. (Bottani, 1998, p. 63, my emphasis)

Taking a less scientifically rigorous approach was seen as a way to prevent too much from being made of the indicators. This policy-orientation meant that the scientists were prepared not to apply the same criteria of validity and soundness as they would, had it been a 'research project'.

Which indicators to include?

Another major challenge was how to contain the number of indicators. Realities always appear to exceed the capacity of the metrics that describe them. Given the range of factors that influence student attainment, the scientists were faced with the tension between being adequately descriptive and adequately usable:

At the first INES General Assembly held in Semmering (Austria) in 1989, an initial list of some 160 indicators was established, but not discussed, everybody being aware that a set of 160 indicators was unmanageable and moreover unacceptable for policy makers. At the second General Assembly in Lugano (Switzerland) in 1991, this figure was reduced to an

‘ideal’ list of some 50 indicators, derived from a conceptual framework which provided guiding principles for the development of a set of international indicators. It took three editions of EAG [Education at a Glance] to come close to the ideal number of indicators. (Bottani, 1996, pp. 280–281)

These challenges encouraged the adoption of a ‘flexible organising framework’ ‘using both *conceptual* and *pragmatic* bases and incorporating policy concerns’ (Bottani, 1994, p. 335, his emphasis). Bottani went on to elaborate:

An approach of this nature seems to be the most appropriate way to organise comparative international data on education for a number of reasons. First, these indicators will be used less for narrow, immediate management purposes and more for global analysis of general trends and differences among countries. Second, the issues important to various countries will vary somewhat, so these cannot be used as an organising principle. Thirdly, this approach encompasses the different intellectual traditions of OECD member countries and draws strength from the synergy of the combined approaches. Finally a systematic conceptual view of education has emerged that seems to have general acceptance and validity. (Bottani, 1994, p. 335)

The hope that the indicators would be used ‘less for narrow, immediate management purposes and more for global analysis of general trends and differences’ appears naïve in hindsight. Once the indicators enter the world and begin to circulate, it becomes very difficult to control how they will be used. Footnotes, cautionary prefaces and technical manuals are quite ineffective in stopping causal inferences from being made. As Latour (1987) has warned, the fate of facts is in the hands of the user.

Scientists lose ground

As a consequence of panic with regard to the quality of education and the quality of data to support policies, following *A Nation at Risk*, ‘the players’ involved in the decisions about education indicators were increasingly policy makers rather than the scientists and researchers:

In the third phase, the players change; they are no longer the specialists in social or education sciences but the policy-makers themselves, those with the responsibility of management and change in education systems. Their appearance on the indicator scene takes place between 1987 and 1992 following the crisis and uncertainty surrounding the quality of teaching and education systems. (Bottani, 2008, p. 16)

The chaotic jumble of priorities and passions of the 1970s and 1980s resolved into a situation where policy makers became the dominant stakeholders, being the prime users – *and funders* – of the indicators. So the report with the new indicators, developed by INES and presented at the ‘international meeting of indicator creators and decision-makers’ (Bottani, 2008, p. 16) in Lugano for discussion, was presented to a meeting where policy makers had a dominant role. Here, after a heated debate, it was decided to include education attainment data into the report, and to publish it as the first EAG (1992):

It was following an intense discussion [between] opposing policy-makers and scientists during the Lugano plenary sessions that the member countries mandated OECD with revising the prototype set of indicators to include a section on student attainment, using the very patchy data available on the international scene at that time. (Bottani, 2008, p. 16)

The Lugano meeting can be seen as a watershed moment in the relations between the scientists and the policy makers. It marked the occasion when scientists started to lose their grip even on the ‘production’ side of the indicators, let alone their ‘use’ side. At the beginning, measurement experts, education experts and social science experts – the scientists – sought input from policy makers to understand what type of data might be useful to them. They translated this into measurement practices as best they could. In devising these indicators, they refrained from insisting on precision and did not quibble too much about validity. But they still had control over how much was compromised. With the 1991 meeting and the forced agreement to include student attainment data, the control appears to have been wrested from the statisticians and social scientists and assessment experts, and placed in the hands of the government officials, with psychometricians as allies. As Bottani elaborates in an article in the *International Journal of Educational Research*:

It is evident that OECD did not find appropriate data for producing a stable set of achievement indicators which provide information expected by the public, users of education systems and policy makers (OECD, 1995). In EAG 3, OECD was obliged to use, for the second time, data from the 1991 IRA Reading Literacy Survey (Elley, 1992). In EAG 2, reading literacy data had been used for calculating three standard indicators on student achievement: multiple comparisons, student score distributions, and within and between school variances. In the third edition, the same data were used for calculating an indicator on the progress in reading between the ages of 9 and 14. The survey’s background questionnaires were used for calculating a second indicator on the amount of reading, based upon the frequency of reading as reported by 14-year-olds. (Bottani, 1996, pp. 283–284)

This use of the same data to make multiple inferences, and using data to provide information for which it is not designed, was anathema to science. This could be discussed in a research journal, but with policy makers playing a more dominant role within the OECD, the scientists were forced to include some very uncertain attainment results. So disgruntled was Bottani with this use of scientifically unacceptable data, that he notes: ‘one can get the impression that these indicators have been invented to avoid leaving empty cells in the subgroup of student achievement’ (Bottani, 1996, p. 284).

The obsession with performance indicators

In the years that followed, the focus on student attainment did not diminish in any way. In the 2000s, which Bottani (2008) describes as a decade of ‘obsession with performance indicators and the triumph of comparative psychometry’, a number of changes were made, including improvements in the quality of data, a modification of the theoretical framework underpinning the indicators, and, perhaps the most significant of all, the launch of PISA, reflecting the rising importance of student attainment indicators.

The most striking result of this phase is the increased importance given to performance indicators. There were 9 of them in the 2001 edition where a distinction was still made between a section dealing with ‘performance at individual and social levels, and on the labour market’ (5 indicators) and student attainment (4); in the 2002 edition, they increased to 14; 15 in 2003; and 12 in 2004. With the 2002 edition, the section concerning the set of ‘performance’ indicators which was the last item in the Education at a Glance index for some ten years, was moved up to take position as the first item. (Bottani, 2008, p. 17)

What started out as a project to provide broad indicators as a useful basis for ‘global analysis of general trends’ rather than ‘narrow, immediate management purposes’ changed dramatically as the outcomes indicators became more and more important and occupied pride of place in EAG. The focus on student assessment brought to the fore another community of experts – psychometricians – who appeared to gain traction with policy makers. The focus on outcome data dramatically changed EAG:

the conceptual framework and organisation of education indicators was changed in 2002, without any truly democratic debate on the subject. All of a sudden, a decade after the publication of the first set of international education indicators which, after much hesitation and precaution, included five performance indicators labelled ‘experimental and temporary’, at the heart of the most renowned set of education indicators in the world, we find performance indicators based on data collected during a mass assessment programme designed to serve the cause and processed using specific psychometric methodologies imposed by an influential scientific community. (Bottani, 2008, p. 18)

The issue for Bottani with the kind of inferences encouraged by the inclusion of such performance data in EAG is that the OECD confuses ‘indicators’ with ‘assessment’. Indicators were intended to merely alert policy makers to potential problems, signalling the need for further assessment of the system and the search for possible remedies. But EAG had now come to present itself as an *assessment of education systems*, combining in a single framework a host of diverse entities from shifts in systems to indicators on education provision to assessment of individual attainment and system efficiency.

The curse of success

The move from the first hesitant steps amidst scepticism about the feasibility of the project, to its confident and strident assertions about all manner of information about education systems was now complete. The first EAG (1992), a bilingual document in French and English, was 150 pages long. The second, 1993, had already doubled in size to 300 pages. By 1995, the French and English versions were published separately and each was 370 pages long (Bottani, 2008). Today it stands at 550 pages, as more and more entities have come within the fold of these OECD indicators.

The obsession with performance data is now closely coupled with a tendency to make a range of causal inferences. The early cautions about using the indicators only as broad-brush descriptions rather than evaluations have been long forgotten. The OECD indicators have now become institutionalised and deeply entrenched in policy making processes in many countries (Smith & Baker, 2001).

The ‘numbers people’ within the OECD continue to be challenged with several aspects of EAG. One of their frustrations relates to how the data are misunderstood, because policy and media people go by numbers without an appreciation of all that lies beneath those numbers. The highly technical nature of EAG serves sometimes to misguide rather than to illuminate, as one OECD official described evocatively:

I think we were numbers people, and we should become policy people. We should change a little bit because EAG was and still is quite a technical publication and now it is no more sufficient to keep this technical part. The technical part is very important for us; you will see if you go through the publication, sometimes the sentence is quite complex because for us ... [the] ways we use these terms is quite complex. [Take] graduation rate – do you know what is ‘graduation rate’? It is very complex, because behind this graduation rate there is a lot of

calculation. The calculation is very complex. People will mix attainment, graduation rate, completion ... they will mix [these things] all the time. For me, [these are] completely different. They come from different sources, there are different definitions behind them, etc. So in this publication you have different concepts which are very technical and very well explained. ... But when it is used for media, it will be '[x%]of Australians have a tertiary degree'. (OECD official, interview transcript)

The EAG continues to see itself as a document in the service of policy makers. It anticipates policy questions and attempts to provide the answers to those questions. While it does not explicitly provide policy advice, it is conscious of its ability to shape the thinking of policy makers and the public through its engagement with them:

We have been asked to answer the questions policy makers might have. We don't give advice, but we are here to use our data for the questions of policy makers. Earlier it was more technical – when we talk of graduation rate, when we compare, when we talk about the finance we compare the finance, but now we should not stay with this type of information ... we should give some figure to the policy maker and give them some direction – to compare countries together to give some direction. (OECD official, interview transcript)

The complexity of the technical is such that the message sometimes needs to be simplified until it is no longer even accurate, in order to be understood:

We have changed our work also. And sometimes we have to forget all the technical issue, because it won't be understood by media, so we know that we have to say some sentences to media that are not correct, because we are statisticians and [we know] what is behind [the numbers], but we should have stories that [are] understandable to everybody. (OECD official, interview transcript)

These changes have gone hand in hand with the growing influence of the OECD on the economic and education policies globally. Their reach has gone far beyond the OECD member nations, as PISA has gained popularity – non-OECD member nations outnumber the members in PISA. OECD data are now widely accepted in a number of spheres, and education has become an important aspect of its work.

The story of EAG provides an opportunity to study closely the tug of war between science and politics, between knowledge and governance and between different types of expertise. But it also demonstrates how the two projects – the scientific project and the governance project – are impossible to keep apart. The politics of science might be different – but it requires the same negotiations and translations and domestications that the arena of policy and governance need. Retaining a sense of technicality and neutrality is important not only to legitimate the indicators, but also the policies that use these indicators as their basis.

Producing calculable worlds

The INES story illustrates that description, from the outset, requires production. No description seems possible without some underlying conceptual framework or theoretical basis, however compromised. Indeed, we might argue that production precedes description – i.e., the world is changed in order to render it describable (Scott, 1998). When the 'world' involved is very diverse and complex, it must first be standardised

using a variety of classificatory mechanisms, mathematical formulae, selections and deletions before any description can be attempted.

Once the measurements are in place and become powerfully routine, as with EAG, the repeated patterns of indicators create certain expectations which form a background against which things appear as expected (Thrift, 2004). The associations made in the indicators – between performance and the GDP, or between per-student expenditure and student performance – become reified as causal or otherwise related entities, even if this was not the original intention. In turn, governments use this information to change their policies. Policy convergence owes much to these comparative tables and charts, which promote associations that might not have been made otherwise. At the same time, what gets left out of these calculations would arguably lose importance and currency in the minds of the public and of policy makers.

For any progress in measurement to be made at all, the first thing that needed to be forged – ‘invented’, to use Bottani’s term – was ‘a new style of co-operation’ between various actors:

Considering the hostility of many educationalists and key decision makers towards the development of education indicators when the activity was started in 1988, it was necessary to invent a new style of co-operation between member countries, able to generate mutual trust and consensus among all participants. (Bottani, 1998, p. 71)

The complexity of education systems, the requirements of large amounts of data from a variety of sources, and the lack of an agreed model or framework meant that agreements had to be constantly negotiated. INES depended on the data produced at national levels to do its international calculations. The idiosyncrasies of national systems meant that data producers also needed to agree to new ways of collecting data or provide ways to translate their data into formats that articulated with INES requirements:

The development of a set of indicators at the international level is a complex, multi-stage operation. Many actors are involved, having various degrees of accountability. Moreover, data gathering in education is not easy: some information is rather simple, some consumes a lot of resources, some may be considered ethically indefensible. It is therefore essential to reach an agreement between the data producers and data providers. (Bottani, 1998, p. 71)

Bottani reports that this emphasis on collaboration and a *participative and democratic approach* was vital to the development of the indicators. But the cooperation came at a cost:

The Networks resemble clubs – a set of people who meet regularly and acquire a ‘group identity’ find it easier to work together than strangers. This makes them more effective, but also potentially leads to decisions taken in the interests of the club that take insufficient account of wider interests and priorities. (Bottani, 1998, p. 73)

The INES ‘General Assemblies’ where the networks came together, along with a range of other stakeholders, provided the forum for debate and discussion. The heated debates of 1991 which resulted in the inclusion of student attainment indicators in the first EAG is an example. However, after 1992, these Assemblies became much tamer affairs, particularly as the participation by policy and government representatives increased:

Between 1989 and 1995, the INES (International Indicators of Education Systems) project organised three General Assemblies with the participation of practically all those concerned worldwide by the production of a set of international education indicators. The last of these events was held in 1995 at Lahti (Finland). Since then there have been no more General Assemblies concerning the international education indicator project. ... OECD organised a fourth General Assembly of the INES project in Tokyo in 2000, but this meeting was a General Assembly in name only, for participation per country was reduced to limited delegations and the indicator producers no longer participated. And so it was an intergovernmental conference by OECD standards which took place in Tokyo rather than a General Assembly. (Bottani, 2008, p. 17)

Over time, the indicators have become taken for granted and routinised, produced year after year. Policy makers have become so used to these indicators that even if the EAG team suggests that some tables or indicators might be dropped, there is resistance to this move. Even when an indicator disappears from the print version, it is still preserved in the web version.

[I]t is very difficult – it's a question mark we have each year – should we drop some of the indicators? But people are used to them, and if we drop one table, you can be sure we will receive some question about why this table has been dropped. It's like – some people say it's a bible. It's not a bible, but people are just used to working with some indicators and they don't like it if we decide to drop them. (Interview transcript, OECD Official)

The consultation with countries still occurs, with meetings twice a year:

So for the organisation of this publication, we work with the countries. We have two meetings per year ... we meet all the countries to discuss the table of contents of EAG 2013 ... we have to decide which indicators we will keep and which we will drop and we try to have for each publication a new story ... Even if we drop a table, it will go to the web, we don't drop it [altogether], because we have the information and it could be useful for [some] people. (Interview transcript, OECD Official)

Consensus can be regarded as a form of translation (Callon, 1986; Latour, 1999) which silences or at least minimises obstruction. By incorporating the varied interests – or by domesticating countries into needing what is on offer rather than what is not – EAG can be routinised and produced with efficiency. The protracted debates have turned into agreements, which in turn have become routines – routines of data generation, applications of statistical formulae and of annual publication. Such methodological routinisation produces a 'mechanical objectivity' which is prized by policy makers (Porter, 1995).

Calculation and critique as moral projects

That measurement is a productive rather than descriptive activity, as I have demonstrated in this paper, has been argued by a number of STS scholars (Jasanoff, 2004; Knorr Cetina, 1981; Latour, 1987; Law, 2009b). There are two aspects to this productivity that are worth teasing out. The more easily argued one is that once a measurement is in place it acts upon the world by changing understandings and behaviours. The other is that investing 'a character of calculability' (Mitchell, 2002) itself changes the world. When we see measurement as not just imperfect descriptions (i.e., reductionist, inaccurate, etc.), but

as *world-making processes*, critiquing them becomes not just an epistemological exercise, but a political and ontological one.

When we regard measurement as a performative or productive process that brings worlds into being, critiquing measurement becomes a form of interference (Law, 2007) which is necessarily a moral project. Jenny Oakes (1986) asserts that ‘we cannot be unaware of the political pressure resulting from the mere existence of a set of indicators’ (Oakes, 1986, cited in Bottani, 2008, p. 15). She explains that teachers and schools might try to influence data in their favour, or influence the choice of indicators, particularly since they often have no control over the many factors for which they could be blamed. Bottani also raises the issue of the performative aspects of the indicators:

Can we really maintain that the use of indicators constitutes a neutral cognitive approach which does not aim to modify education systems? ... Is it not rather naive to claim that indicators are mere tools for acquiring knowledge with no incidence on the realities of education, on how the education system performs? Is the fact of producing a set of indicators, supported and encouraged by educational authorities, completely neutral? (Bottani, 2008, p. 21)

Bottani also makes the more interesting point of that there is no ontological divide between reality and its modification, since knowing involves changing:

... no cognitive approach leaves the subject of its observation, measurement, analysis, comparison, deciphering, completely unscathed. There is no clear ontological divide between science and techniques, between cognitive reality and its modification. It is therefore impossible to attribute a distinct ethical status to each of these two elements and consider that we have unlimited freedom when it comes to cognitive processes and a different, limited responsibility when we apply knowledge to processes or acts aiming to modify reality. Indicators produce knowledge of education systems and the moment we design, implement or produce indicators, they modify systems. (Bottani, 2008, p. 22)

This assertion from Bottani perhaps explains why some of the people most involved with the project of indicator production have felt compelled to write and speak with such frankness about the precarious and tentative nature of these indicators and why they continue to challenge and raise cautions about indicators (see, for example, Bottani, 1996, 1998, 2008; Heyneman, 1999; Oakes, 1986; Sauvageot, 2008; Smyth, 2008).

Bottani’s assertion that to know is to change, and his challenge that there is no ontological divide between ‘science and techniques, between cognitive reality and its modification’ sums up the moral imperative of my own project of doing ‘sociologies of measurement’, at a time when such measurements have become deeply influential in policy. No accounts of contemporary policy enactments can be given today without an account of various forms of standardisations, classifications and quantifications and other means of translating the world into numbers (Gorur, 2011). While the role of numbers in policy is well recognised, critique is often content to either show that numbers are reductionist or to challenge their accuracy. While both of these are important, neither, I argue, is sufficient. Those critiques arise from a representationalist, rather than productive or performative, understanding of measurement (Pickering, 1995; Woolgar, 1991a). Detailed accounts of the making of numbers and their insinuation in policies are required to understand their instrumentalism and their world-making functions (Woolgar, 1991a). The former – an understanding of their instrumentalism – provides potential points of

active intervention and challenge. The latter – an understanding of measurement as a productive rather than a descriptive practice, provides the moral and ethical impetus to take on such studies.

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